WASTEWISE SCAN · TRACEABILITY MATRIX

WasteWise Scan traceability matrix

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wastewise-scan		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

WasteWise Scan traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 17 user requirements, 17 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	Identify common resins	WSC-CAL-001, WSC-BLD-001, WSC-STM-001, WSC-OPT-001	Analysis, paper band study WSC-CAL-001 B4 to B7 (TRL 3); labelled item set of 200 or more (TRL 4)	At risk: 89.5 % with eight bands at 1.5 % band scatter, 96.7 % at 1.0 %; PE against PP 93 %; 1,650 nm band centroid 1,628 nm [B2, B4 to B7]
UR-002	PR-002	Fast result	WSC-CAL-001, WSC-BLD-001, WSC-STM-001	Analysis, timing budget WSC-CAL-001 E (TRL 3); timed at TRL 4	Met on paper: 0.64 s with the four-cycle sequence of fix F2 [E2]
UR-003	PR-003	Fail safe on unreadable items	WSC-CAL-001, WSC-STM-001	Analysis of the black threshold WSC-CAL-001 C6 (TRL 3); confidence threshold study on items (TRL 4)	To verify at TRL 4: black items fall well below the 0.08 reflectance threshold [C6]; the wrong-result rate needs item data
UR-004	PR-004	Work in sunlight	WSC-CAL-001, WSC-BLD-001	Analysis, stray light WSC-CAL-001 D (TRL 3); indoor and sun test (TRL 4)	Met on paper (estimate): 0.08 % on opaque items, 0.21 % on backed clear items with fix F2; unbacked clear items refused [D5 to D7]; rests on the assumed cap leak and ambient change (STEM S5)
UR-005	PR-005	Last a shift	WSC-CAL-001, WSC-BLD-001	Analysis, power budget WSC-CAL-001 F (TRL 3); measured at TRL 4	Met on paper: 21.9 h [F2]
UR-006	PR-006	One-handed and light	WSC-CAL-001, WSC-BLD-001	Analysis from the model volumes WSC-CAL-001 A (TRL 3); weighing at TRL 4 (WSC-DEC-001)	Not met: 239 g, 259 g with the cap (9 g over); body 160 x 62 x 34 mm [A1, A3]
UR-007	PR-007	Readable by anyone	WSC-CAL-001, WSC-BLD-001, WSC-STM-001	Analysis, contrast WSC-CAL-001 I (TRL 3); design review with users	At risk: about 5.2:1 where the sun hood shades the screen; about 1.6:1 with the sun near the screen normal or over the open side [I3, I4]
UR-008	PR-008	Local price grade	WSC-CAL-001	Inspection of the design (TRL 3); design review with the cooperative	Met by design
UR-009, UR-017	PR-009	Log scans	WSC-CAL-001	Analysis, storage WSC-CAL-001 G (TRL 3)	Met on paper: about 131,000 scans in an 8 MB partition [G1]
UR-010	PR-010	Rugged	WSC-CAL-001, WSC-BLD-001, BOM	Analysis of drop loads WSC-CAL-001 J (TRL 3); drop and spray tests (TRL 4)	To verify at TRL 4: corner drop about 1,200 g mean on an assumed 1 mm crush; the 11 x 10 mm cell strap (fix F3) carries the 869 N peak with a margin of 1.45 [J1, J2, J4]

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-011	PR-011	Stay calibrated	WSC-CAL-001, WSC-BLD-001	Inspection of the design and WSC-CAL-001 J3 (TRL 3)	Met by design: 5 % drift after about 12.5 C of change [J3]
UR-012	PR-012	Low cost	WSC-CAL-001	Priced BOM, WSC-CAL-001 K (TRL 3)	Over target: USD 167.50, USD 17.50 over the USD 150 value-engineering target [K2]
UR-013, UR-017	PR-013	Work with WasteWise-ml	WSC-CAL-001	Inspection, interface note with WasteWise-ml	At risk: record decided on this side (WSC-DDR-002 v0.4), not yet adopted by WasteWise-ml
UR-014	PR-014	Eye-safe illumination	WSC-CAL-001, WSC-BLD-001	Analysis WSC-CAL-001 H (TRL 3); datasheet values and measurement (TRL 4)	Met on paper (estimate): 3.9 W/m2 at 200 mm; retinal 3.8 % of the limit [H2, H3]
UR-015	PR-015	Safe cell	WSC-CAL-001, WSC-STM-001	Analysis WSC-CAL-001 J2, J4 (TRL 3); STEM S10, S17, S18; drop test at TRL 4	To verify at TRL 4: the 11 x 10 mm strap (fix F3) carries 869 N at a bending margin of 1.45, on an assumed crush [J4]
UR-016	PR-016	Buildable by a local maker	WSC-CAL-001, WSC-BLD-001, WSC-STM-001	Inspection of the build plan WSC-BLD-001 and the BOM (TRL 3)	Met by design (WSC-BLD-001; the 8.4 mm hole in the LED window of fix F1 needs a drill and an 8 mm diamond core drill)
UR-001, UR-011	PR-017	Optical crosstalk	WSC-CAL-001, WSC-BLD-001, WSC-STM-001	Analysis WSC-CAL-001 C5, C7, C8 and STEM S12 (TRL 3); wall-opacity coupon, open-air and dirty-window readings (TRL 4)	To verify at TRL 4: 0.003 % with the baffle through the glass on assumed leak values (fix F1; 10 % before the fix); the tube wall needs an opacity of 2 x 10^{-2} or better [C8]

Gaps

None.